Roll Number: 2024247

B.E. I (Computer Engg. Section A and B) Regular Examination January 2024

1ACRC2 Applied Chemistry & Environmental Science

Duration: 3 Hours **Max Marks: 60**

Note: Answer any Two sub-parts of a given question carrying equal marks.
Give Chemical equations, labeled diagrams and examples wherever needed.
Different parts of a question should be attempted in continuation.
Neat presentation is desirable.

Q.1A Define Titration. Explain roles of EDTA, EBT and metal-ligand complex formation in water hardness determination titration. 20 ml. of hard water require 12 ml. of N/100 EDTA solution. Calculate total hardness of water. Is this water fit for drinking? 6

B List and explain various problems associated with using hard water in boilers. Suggest ways to avoid these problems. 6

C Explain chemistry, process and working of any one external water softening method including its advantages and limitations. 6

Q.2 A Explain the term Engineering Materials. Describe classification of engineering materials including their properties and examples. 6

B Describe industrial manufacturing of Ordinary Portland Cement. 6

C Why and where Refractory materials are used? What factors are to be considered for selection of a Refractory? 6

Q.3 A Define Lubricants. How lubricants work? Explain any one mechanism of lubrication in detail. 6

B Describe classification of lubricating oils with examples. What useful properties are required in a good lubricating oil? 6

C Write Notes on (Any two): 1. Redwood Viscometers 2. Greases 3. Cutting fluids 6

Q.4 A Explain Beer-Lambert's Law and its applications. A $3*10^{-4}$ M solution of Aniline has A = 0.504 at 280 nm, when measured in a 1.00 cm cell. Find A and %T of $1.5*10^{-4}$ M solution of Aniline when measured at same λ but in a 0.5 cm cell. 6

B Define and classify Spectroscopic techniques. What is the basis (principle) of qualitative and quantitative analysis by spectroscopic method. 6

C Write explanatory notes on (Any two): 6
1. Molecular finger prints 2. Column Chromatography 3. Colorimeter

Q.5 A Name various segments of Environment. List and explain various adverse impacts of development on our Environment. 6

B Explain water pollution including list of pollutants, sources, harmful effects, control measures. 6

C Write explanatory notes on: (Any two) 6
1. Eutrophication 2. Acid Rains 3. Sustainable development